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Problem 1:

$M_{2,2}(\mathbb{C}) \times M_{2,2}(\mathbb{C}) \to \mathbb{C}$

$$\langle A, B \rangle = tr(A^* B)$$

To show that it is an inner product:

- → Conjugate Symmetry
- → Linearity in first argument
- → Positive definiteness

Conjugate Symmetry: $\langle A, B \rangle = \overline{\langle B, A \rangle}$

$$\langle A, B \rangle = tr(A^* B) \quad \to (i)$$

$$\langle B, A \rangle = tr(B^* A)$$

$$\overline{\langle B, A \rangle} = \overline{tr(B^* A)}$$

$$tr(X) = \overline{tr(X^*)}$$

$$\overline{\langle B, A \rangle} = tr((B^* A)^*)$$

$$\overline{\langle B, A \rangle} = tr(A^* B)$$

from (i)

$$\langle A, B \rangle = \overline{\langle B, A \rangle}$$

linearity: $\langle \alpha A + \beta C, B \rangle = \alpha \langle A, B \rangle + \beta \langle C, B \rangle$

$$\begin{aligned}
\langle \alpha A + \beta C, B \rangle &= tr((\alpha A + \beta C)^* . B) \\
&= tr((\alpha A^* + \beta C^*) B) \\
&= \alpha\, tr(A^* B) + \beta\, tr(C^* B) \\
&= \alpha \langle A, B \rangle + \beta \langle C, B \rangle
\end{aligned}$$

Proofed

Positive definiteness: $\langle A, A \rangle > 0$

$\langle A, A \rangle = 0$ iff $A = 0$

$\langle A, A \rangle = tr(A^* A)$

A^*A will produce a positive matrix therefore

$\langle A, A \rangle > 0$

$A^*A = 0$ only if $A = 0$

$\langle A, A \rangle = 0$ iff $A = 0$

because zero matrix is the only matrix whose conjugate transpose $\times$ by itself is zero.

Problem 2:

Orthonormal if

$\rightarrow \langle v, w \rangle = 0$

$\rightarrow \|v\| = 1$

let

$A_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\quad A_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

$A_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\quad A_4 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$\langle A_1, A_2 \rangle = \frac{1}{2} tr\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right) = \frac{1}{2} tr \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 0$

$\langle A_1, A_3 \rangle = \frac{1}{2} tr \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = 0$

$\langle A_1, A_4 \rangle = \frac{1}{2} tr \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = 0$

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$$\langle A_2, A_3\rangle = \frac{1}{2}\,tr\begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = 0$$

$$\langle A_2, A_4\rangle = \frac{1}{2}\,tr\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = 0$$

$$\langle A_3, A_4\rangle = \frac{1}{2}\,tr\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} = 0$$

Normalization :-

$$\langle A_1, A_1\rangle = \frac{1}{2}\,tr\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{2}\,tr\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$$

$$\langle A_2, A_2\rangle = \frac{1}{2}\,tr\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \frac{1}{2}\,tr\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$$

$$\langle A_3, A_3\rangle = \frac{1}{2}\,tr\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \frac{1}{2}\,tr\begin{bmatrix} -i^2 & 0 \\ 0 & -i^2 \end{bmatrix} = 1$$

$$\langle A_4, A_4\rangle = \frac{1}{2}\,tr\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \frac{1}{2}\,tr\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$$

Therefore the basis are orthonormal.

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Problem 8:

$[A]_B = (\langle B_1, A\rangle, \langle B_2, A\rangle, \langle B_3, A\rangle, \langle B_4, A\rangle)$

$$\langle B_1, A\rangle = \frac{1}{\sqrt{2}} tr\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 5 & 2-3i \\ 2+3i & -3 \end{bmatrix}\right)$$
$$= \frac{1}{\sqrt{2}} tr\begin{bmatrix} 5 & 2-3i \\ 2+3i & -3 \end{bmatrix} = \sqrt{2}$$

$$\langle B_2, A\rangle = \frac{1}{\sqrt{2}} tr\left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 5 & 2-3i \\ 2+3i & -3 \end{bmatrix}\right)$$
$$= \frac{1}{\sqrt{2}} tr\begin{bmatrix} 2+3i & -3 \\ 5 & 2-3i \end{bmatrix}$$
$$= \frac{2.2}{\sqrt{2}} = \frac{\sqrt{2}\cancel{\sqrt{2}}.2}{\cancel{\sqrt{2}}} = 2\sqrt{2}$$

$$\langle B_3, A\rangle = \frac{1}{\sqrt{2}} tr\left(\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} 5 & +2-3i \\ 2+3i & -3 \end{bmatrix}\right)$$
$$= \frac{1}{\sqrt{2}} tr\begin{bmatrix} -2i-3i^2 & 3i \\ 5i & 2i-3i^2 \end{bmatrix}$$
$$= \frac{1}{\sqrt{2}} tr\begin{bmatrix} 3-2i & 3i \\ 5i & 3+2i \end{bmatrix} = \frac{6}{\sqrt{2}} = 3\sqrt{2}$$

$$\langle B_4, A\rangle = \frac{1}{\sqrt{2}} tr\left(\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 5 & 2-3i \\ 2+3i & -3 \end{bmatrix}\right)$$
$$= \frac{1}{\sqrt{2}} tr\begin{bmatrix} 5 & 2-3i \\ -2-3i & 3 \end{bmatrix} = \frac{8}{\sqrt{2}} = 4\sqrt{2}$$

$$[A]_B = \begin{bmatrix} \sqrt{2} \\ 2\sqrt{2} \\ 3\sqrt{2} \\ 4\sqrt{2} \end{bmatrix}$$